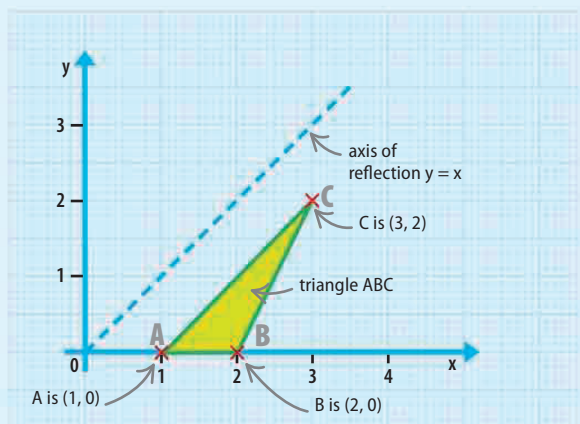
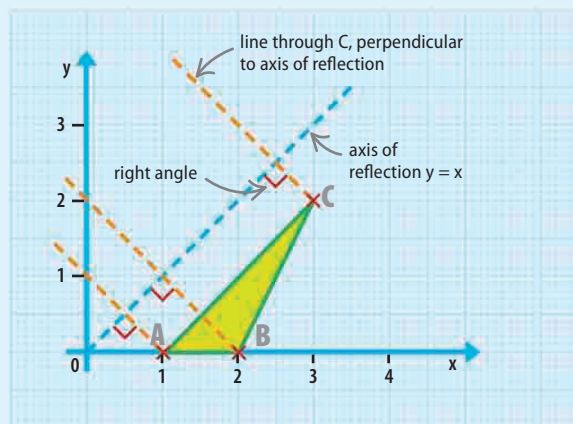


Constructing a reflection

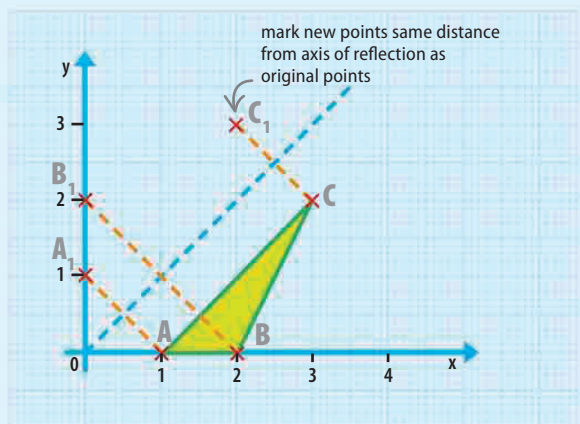
To construct the reflection of an object it is necessary to know the position of the axis of reflection and the position of the object. Each point on the reflection will be the same distance from the axis of reflection as its corresponding point on the original. Here, the reflection of triangle ABC is drawn for the axis of reflection $y = x$ (which means that each point on the axis has the same x and y coordinates).



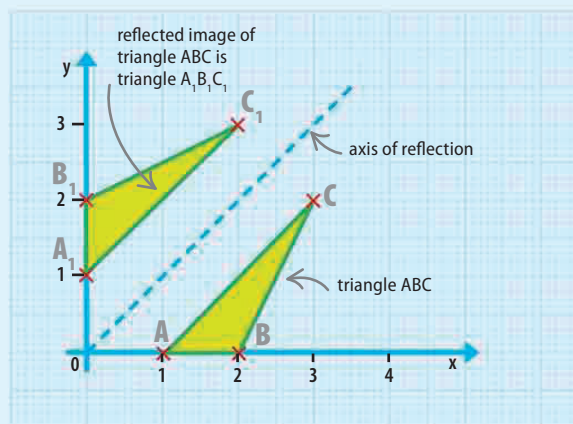
First, draw the axis of reflection. As $y = x$, this axis line crosses through the points (0, 0), (1, 1), (2, 2), (3, 3), and so on. Then draw in the object that is to be reflected—the triangle ABC, which has the coordinates (1, 0), (2, 0), and (3, 2). In each set of coordinates, the first number is the x value, and the second number is the y value.



Second, draw lines from each point of the triangle ABC that are at right-angles (90°) to the axis of reflection. These lines should cross the axis of reflection and continue onward, as the new coordinates for the reflected image will be measured along them.



Third, measure the distance from each of the original points to the axis of reflection, then measure the same distance on the other side of the axis to find the positions of the new points. Mark each of the new points with the letter it reflects, followed by a small 1, for example A_1 .



Finally, join the points A_1 , B_1 , and C_1 to complete the image. Each of the points of the triangle has a mirror image across the axis of reflection. Each point on the original triangle is an equal distance from the axis of reflection as its reflected point.